

Manas Pal

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PROFESSIONAL SUMMARY

Backend Engineer specializing in retail-media and ad-tech infrastructure — ad-serving, monetization pipelines, and real-time ranking at scale. 1+ years shipping production features for a retail-media operating system serving 100s of millions of ad requests, with measurable impact on CTR, ROAS, and revenue.

PROFESSIONAL EXPERIENCE

Osmos by Onlinesales.ai

May 2025 – Present | Pune, India

Software Engineer — Performance Engineering

- Architected performance-focused backend services and ETL pipelines powering **retail-media monetization and ad-serving infrastructure** on a platform serving **100s of millions of ad requests** across multiple marketplaces, sustaining **P95 latency of 100–150ms** at peak traffic.
- Designed and shipped a new **keyword-to-category prediction pipeline** with a fallback path and redesigned Redis storage layout — **+0.5 CTR** on the pilot client with no RR loss, and **~20% Redis memory reduction** across all clients on rollout.
- Led the Performance team's portion of a cross-team **Redshift → BigQuery migration** for critical pipelines: rewrote queries for BigQuery-native execution, refactored ETL workflows, and shipped an automated parity-validation framework diffing on primary keys/emails; in the process, identified and fixed/optimized legacy code issues, driving **up to 20%** lower pipeline execution time and cloud cost.
- Built a **budget-pacing algorithm** — including for a new campaign-group (CG) hierarchy layer introduced to support a major enterprise client (PayPal) onboarding — that analyzes real-time traffic trends to throttle spend and prevent budget exhaustion during peak hours, working correctly across both single-campaign and grouped-campaign configurations.
- Built, solo end-to-end, automated monitoring frameworks spanning ETL validation, API monitoring, QA regression, and **low-level business-logic anomaly detection** — catching silent failures invisible to standard service/data monitors (e.g. undersized models, malformed or out-of-range inputs, logically incorrect algorithm output) — cutting RCA turnaround for leadership escalations by **~50%**; rotate through production on-call, triaging pipeline failures and driving RCAs.
- Leveraged AI-assisted development tools (ChatGPT and Claude) to accelerate debugging, SQL optimization, code reviews, documentation, test-case generation, and solution exploration, improving engineering productivity while ensuring all production code was validated through rigorous testing and peer review.
- Mentor interns and junior engineers on backend development, debugging, and system design.

Stack: Python, Node.js, Java, Ruby, SQL, BigQuery, Redshift, GCP, AWS, Microservices, Solr, Redis, MongoDB, Git, Linux, OLAP, OLTP, ETL

Osmos by Onlinesales.ai

June 2024 – April 2025 | Pune, India

Started on Infrastructure, transitioned to Performance Engineering

- Optimized the BigQuery-to-BigQuery fetch-and-store workflow with a **MERGE-based strategy** in a hybrid Redshift–BigQuery ETL pipeline — **~40% lower BigQuery compute costs** and **~50% faster ETL execution**.
- Modified a legacy brand-extraction pipeline with new extraction logic and a real-time fallback path — **2× brand coverage, +0.2 avg CTR** across major clients, while holding **P95 latency at 100–150ms** in a latency-sensitive production service.
- Designed and shipped **Node.js and Python microservices** powering internal data-processing workflows and APIs, and built **automated job-orchestration and monitoring** for ETL pipelines across BigQuery and Redshift — improving pipeline reliability and cutting manual intervention.

PROJECTS

Hate Speech Detection in Hindi Language

Oct 2023 – Dec 2023

- Fine-tuned DistilBERT on the HASOC dataset for Hindi social-media sentiment classification, achieving **84% accuracy**; built training pipeline in PyTorch and served predictions through a React front-end.

Aircraft Blade Defect Inspection (Bharat Forge)

Jan 2023 – Jan 2024

- Annotated 3,000+ defects across 500 images and benchmarked detection architectures; selected Mask R-CNN (mAP 0.55–0.59), meeting industry benchmarks.
- Built Flask inference APIs and a React inspection UI with integrated camera SDK (zoom, autofocus, capture).

SKILLS

Domain Expertise: Retail Media, Ad-Serving, Programmatic Advertising, Real-Time Bidding, Monetization Infrastructure, Ranking Algorithms

Languages: Python, Java, JavaScript, C++, PHP, Ruby, SQL, Bash, HTML, CSS

Backend & Frameworks: Node.js, Flask, FastAPI, REST APIs, Microservices

Databases & Data Platforms: MySQL, MongoDB, Redis, Apache Solr, Google BigQuery, Amazon Redshift

Data Engineering: ETL Pipelines, Data Warehousing, OLAP, OLTP

Cloud & DevOps: AWS, GCP, Docker, Kubernetes, CI/CD, Git, Linux

ML/AI: Machine Learning, Deep Learning, NLP | **AI-Assisted Dev Tools:** Claude, ChatGPT, Gemini, GitHub Copilot

EDUCATION

Dr. Vishwanath Karad MIT World Peace University

2021 – 2025 | Pune, India

B.Tech, Computer Science and Engineering — 9.02 CGPA

CERTIFICATES

Machine Learning Specialization (DeepLearning.AI, Stanford) • Responsive Web Design (freeCodeCamp) • AI Job Simulation (Cognizant, via Forage)